

WORKING DRAFT — NOT FOR USE

Rolling Backdrop System

Illustrated Construction and Field Operations Manual

Version	Date	Status	One-line description
Release v0 (unreleased) — Development build D1	2026-09-01	WORKING DRAFT — NOT FOR USE	Standardized master manual with 2026 CBA competition rules, 3/4" wing ballast posts, handle sandbag specifications, and instrument tote guidelines
Release v0 (unreleased) — Development build D2	2026-09-10	WORKING DRAFT — NOT FOR USE	Add detailed vinyl installation procedure, specifications table, hardware relief slits, and snap clamp fastening
Release v0 (unreleased) — Development build D3	2026-09-10	WORKING DRAFT — NOT FOR USE	Add engineered wind relief slit specifications, tiered 15-lb handle sandbag ballast schedule, rear-rail leverage optimization, and high-wind emergency protocol

Intro / Overview

This master manual is organized as two independently printable appendices under one controlled revision history. The complete package contains the system overview, field operations procedures, 2026 competition rules compliance, construction instructions, and purchasing index for the Pine Creek High School Marching Band rolling backdrop prop.



Figure 2. Backdrop units deployed in performance formation on athletic field.

Appendix A: Operations Manual

Covers field transport, entrance and exit procedures, student and parent volunteer training drills, 3/4-inch wing ballast retention posts, double-bagged sandbag requirements, student instrument stowage, 2026 CBA competition prop rules, and post-use teardown/trailer storage. Intended for parent field crew, prop handlers, and student stage crew.

Appendix B: Construction Manual

Covers workshop safety, materials, 2x4 rolling cart base fabrication, decking and 3/4-inch ballast post flange installation, 1-5/8 in. steel pipe upright frame assembly, diagonal support strut pre-drilling and mounting, vinyl banner installation with snap clamps, inspection checklist, and source lookup. Intended for fabrication volunteers and workshop build teams.

Printing and separation

Each appendix begins with an explicit cover page and uses independent page numbering (labeled A- for operations and B- for construction). When printing work packets for rehearsal or fabrication sessions, print the relevant appendix in its entirety, including its cover sheet.

Acknowledgments and Design Credit

Special thanks and credit to the Plainfield North Bands for creating the original rolling cart design and foundational engineering concept. The Pine Creek High School Marching Band adapted and refined this design for our field operations, 3/4" vertical wing ballast retention posts, turf ballasting standards, and custom 3D-printed alignment fixtures.

PART OF THE ROLLING BACKDROP CONSTRUCTION AND FIELD OPERATIONS MANUAL

APPENDIX A

Operations Manual

Field transport, entrance/exit, training, ballasting posts, instrument stowage, 2026 rules, and storage

This appendix is designed as a complete, independently printable work packet which can be distributed separately to parent volunteers, student stage crew, and field handlers.

Audience: Field operators, parent prop crew, and student handlers

Contents: Transport & field entrance; student/volunteer training; wing ballast posts; double-bagging rules; student instrument stowage; 2026 CBA rules; trailer storage

Print or carry: All pages labeled A-

Parent revision: Release v0 (unreleased) | Development build D3 | September 10, 2026

When separated from the master document, keep this cover as the first page of Appendix A.

Field Operations

1.1 Roles and Crew Organization

Field execution of the rolling backdrop props relies on a coordinated division of responsibility between parent volunteers and student performers:

A. Parent Pit & Logistics Crew:

1. • **Trailer Unloading & Staging:** Offload backdrop carts from the equipment trailer and roll them to the stadium entrance tunnel or staging chute (Figure 3).
- **Ballast Loading & Staging:** Assess stadium wind conditions and stage double-bagged 15-lb sandbags according to the quick-reference schedule below (see Section 1.4 for full engineering analysis and leverage details):

Wind Condition	Wing Posts (3/4" Pipe)	Rear Rail (2x4 A)	Total Ballast	Field Status
Calm / Normal (0–12 mph)	4x 15-lb bags (2 per wing post)	None	60 lbs (213 lb prop)	Standard Deployment
Advisory (12–18 mph)	6x 15-lb bags (3 per wing post)	None	90 lbs (243 lb prop)	Attentive Staging
High-Wind (18–22 mph)	6x 15-lb bags (3 per wing post)	3x 15-lb bags laid flat	135 lbs (288 lb prop)	Max Safe Turf Limit
Abort (>20 mph / gusts >25)	—	—	—	NO-GO / LAY FLAT

- **Pre-Show Safety Check:** Verify all four swivel casters rotate freely, 360° swivel action is smooth without binding, mounting hex nuts are tightened securely, and strut hardware has zero play.
 - 2. • **Staging Order:** Arrange backdrops in numerical show drill order at the field gate.
 - 3. • **Exit Catch Crew:** Position parent volunteers (holding official Field Pass wristbands per Rule 9.07) at the stadium exit gate to receive carts from students, assist with deceleration, remove ballast sandbags, and pack the trailer.
- #### B. Student Prop Handlers / Performers:
4. • **Field Operators:** Students have sole responsibility for pushing and maneuvering the backdrop carts onto the field, executing show transitions, and pushing them off the field at the conclusion of the performance.
 5. • **Crew Pairing:** Exactly two student handlers are assigned to each backdrop cart, positioned on opposite outer end perimeter rails (B).

6. • **Instrument Stowage & Future Cradle Revision:** The central opening cradles an HDX 14-gallon tough storage tote (Index [9]) for student handlers to temporarily place their musical instruments while pushing carts onto and off the field. Operational Note: The 14-gallon tote accommodates smaller instruments (flutes, clarinets, trumpets, alto saxophones), but is constrained when handlers play larger instruments (mellophones, trombones, baritones, tenor/bari saxes, or battery percussion). A future design revision is planned to engineer an expanded modular instrument cradle system.
- **On-Field Positioning & Ground Chocking:** Carts are equipped with non-locking swivel casters to maximize rolling agility during rapid field entrances and exits. When props reach their drill coordinates in windy conditions, handlers or parent pit crew deploy supplemental double-bagged 15-lb sandbags on the ground directly against the wheels to serve as wheel chocks, preventing field drift during the performance.



Figure 3. Assembled rolling backdrops staged and ready for field deployment.

1.2 Field Entrance, Positioning, and Egress

Timed performance intervals require strict adherence to navigation protocols:

- 1. Stadium Gate to Field Transit:** Students take control of the staged carts at the field gate, placing their instruments safely inside the central tote bin. Push carts in two-person teams with the upright frame facing perpendicular to travel to maximize forward visibility.
- 2. Track & Curb Crossovers:** Approach all running track crossings, track protector mats, and turf curb transitions square-on at a controlled walking pace. Both front casters **MUST** cross curbs simultaneously. Never approach curbs at an angle, which can bind casters or induce frame tipping.
- 3. Drill Coordinate Placement & Ground Chocking:** Guide the cart to its exact field drill coordinate. Because carts utilize non-locking swivel casters, handlers position the cart square to the front sideline. When needed in breezy conditions, place extra double-bagged 15-lb sandbags on the ground firmly against the caster wheels as temporary wheel chocks to eliminate roll drift, retrieve instruments from the center tote, and take show positions.
- 4. Rapid Field Egress:** On the final performance cue, handlers place instruments in the tote, quickly lift any ground sandbag chocks onto the cart base or hand them to sideline crew, and push the carts in assigned exit order directly toward the exit gate. All props must maintain continuous movement off the field per Rule 8.05.

1.3 Student and Volunteer Training Protocol

All student handlers and parent volunteers must complete practical training before competition deployment:

- 1. Proper Hand Placement & Leverage Safety:** Handlers must ALWAYS keep hands low, grasping the outer wooden 2x4 perimeter rails (B) or inner reinforcement joists. NEVER push, pull, or apply body weight to the upper 1-5/8 in. steel pipe frame or diagonal struts. Pushing high creates massive overturning leverage that can tip the prop or bend steel mounting brackets.
- 2. Instrument Transition Drills:** Practice smooth, rapid instrument placement and retrieval from the central carrier tote during timed entry, staging, and exit drills.
- 3. Rehearsal Run-Throughs:** Conduct full-speed timed entry, positioning, ground sandbag chocking, and egress drills during dress rehearsals on both synthetic turf and natural grass surfaces.
- 4. Handshake & Communication Protocol:** The student on the outer sideline edge acts as the primary caller ('Clear left', 'Approaching curb', 'On coordinate', 'Chocks set', 'Chocks clear'). Handlers maintain continuous verbal communication during moves.
- 5. High Wind Emergency Response & Abort Protocol:** If sustained winds exceed 20 mph or sudden gusts exceed 25 mph, rolling backdrops MUST NOT enter the performance field (strict safety NO-GO threshold). If severe gusts strike while props are already on the field, student handlers position themselves on the upwind side of the cart, place ground sandbag chocks firmly against wheels, and brace the lower 2x4 framing. At no time should students attempt to hold backdrops from the upper steel pipe rails or climb the frame. Designated parent volunteers with Field Pass wristbands (Rule 9.07) will step onto the field to provide perimeter stabilization or assist handlers in immediately tipping the props flat onto the turf if ordered by band directors.

1.4 Ballasting System, Tiered Wind Schedule, and Double-Bagging Protocol

To maintain vertical stability against outdoor wind shear while strictly protecting stadium athletic turf from rutting and sand contamination, all backdrop props must adhere to this engineering ballasting standard and operational loading schedule.

A. Standardized Tiered Ballasting & Wind Speed Schedule

Ballasting requirements are scaled dynamically based on observed stadium wind velocity and peak gust forecasts prior to field gate ingress:

Tier / State	Wind Velocity	Wing Posts (3/4" Pipe)	Rear Rail (2x4 A)	Total Ballast	Max Safe Gust & Protocol
Tier 1: Normal	Calm to light breeze (0–12)	4x 15-lb bags (2 per wing post;	None	60 lbs (213 lb	16.9 mph max gust. Standard operations;

Tier / State	Wind Velocity	Wing Posts (3/4" Pipe)	Rear Rail (2x4 A)	Total Ballast	Max Safe Gust & Protocol
	mph)	30 lb/side)		total)	zero turf compaction.
Tier 2: Advisory	Moderate breeze (12–18 mph)	6x 15-lb bags (3 per wing post; 45 lb/side)	None	90 lbs (243 lb total)	18.4 mph max gust. Handlers deploy ground sandbag chocks if breezy.
Tier 3: High-Wind	High wind advisory (18–22 mph)	6x 15-lb bags (3 per wing post; 45 lb/side)	3x 15-lb bags laid flat along rear rail A	135 lbs (288 lb total)	22.5 mph max gust. Symmetrical stability; max safe turf limit.
Tier 4: Safety Abort	Severe gale (>20 mph sustained / >25 mph gusts)	—	—	—	UNSAFE. Field withdrawal / NO-GO. Props kept off field or laid flat.

B. High-Wind Rear-Rail Ballast Leverage Optimization

Because the 10 ft x 8 ft vertical upright steel frame sits on the front perimeter rail (A) of the cart base, the structure possesses an inherent tipping asymmetry:

- 1. Critical Failure Mode (Forward Tipping):** When wind blows from the rear toward the front, the cart pivots about the front caster wheels. Because the 48-lb steel frame and vinyl banner sit directly above the front caster line, their deadweight provides zero restoring leverage. As a result, an unballasted prop tips forward at just 12.4 mph (11.2 mph at sea level).
- 2. The Rear-Rail Mechanical Advantage:** The 3/4-in. wing pipe posts sit at mid-depth (22.25 in. from front), providing a restoring lever arm of 1.71 ft to the front casters. By contrast, placing supplemental sandbags along the rear 2x4 framing rail (A) behind the instrument tote increases the lever arm to 3.46 ft (41.5 in.)—more than double the mechanical leverage! Three 15-lb bags placed across the rear rail produce the overturning resistance of six bags stacked on the center posts, equalizing forward and backward tipping thresholds to 22.5 mph.
- 3. Turf Compaction Ceiling:** Total supplemental ballast must never exceed 140 lbs (293 lbs total prop deadweight). Concentrating more than ~75 lbs per caster wheel causes casters to sink into artificial turf rubber infill, multiplying rolling drag and risking permanent turf depressions or track damage.

C. Mandatory Double-Bagging Protocol (CBA Rule 8.05)

All 15-lb sandbags MUST utilize heavy-duty inner plastic liner insert bags (Index [14]) sealed securely inside the outer fabric cordura bags. Under no circumstances may unbagged or single-layer sandbags be brought onto any competition turf. If an outer fabric bag abrades, splits, or suffers seam failure, the heavy-duty inner liner guarantees zero sand escapes onto the performance field.

D. Emergency Reserve Protocol (70-lb Tube Sand)

In the event that 15-lb handle sandbags are depleted or extreme wind gusts threaten during stationary staging, crew may deploy 1 to 2 70-lb Sakrete traction tube sand bags (Index [15]) positioned flat across the rear lumber framing rail. MANDATORY: Every 70-lb tube sandbag MUST be completely wrapped and sealed inside a thick contractor-grade garbage bag to satisfy the secondary containment rule before entering the stadium.

CRITICAL TURF OPERATION & DOUBLE-BAGGING WARNING: Under no circumstances may unbagged or single-layer sandbags be brought onto any competition turf (see Section 1.5, Rule 8.05). In addition, do not exceed 135–140 lbs total ballast under any operational conditions, as excessive weight causes casters to sink into artificial turf rubber infill, multiplying rolling friction and making carts sluggish for student handlers.

1.5 Applicable 2026 Marching Band Competition & Prop Rules

The following governing rules are extracted directly from the official 2026 Colorado Bandmasters Association (CBA) Marching Band Rulebook (Index [17]). All prop builders, parent crew, and student handlers must adhere to these standards:

ANNUAL RULEBOOK NOTICE: *This section reflects the governing rules for the 2026 competitive season. Logistics staff and build leads must review this section annually against updated CBA and Bands of America (BOA) rulebooks to verify continuing compliance.*

Rule 8.05: Props, Equipment, Surface Protection, and Double-Bagging

7. • **Gate Ingress:** All props and equipment must be designed and of a quantity such that they can be brought onto the Performance Field from the band entrance gate.
8. • **Continuous Movement:** Following the end of the band's Performance, all props and equipment must be in continuous movement until entirely removed from the stadium.
9. • **Turf Contact Protection:** All wooden props must be protected with a heavy-duty sustainable plastic product (PVC, Melamine, etc.) where said prop comes in contact with the field surface. Steel props with smooth edges are acceptable without the need for additional protection.
10. • **Secondary Containment for Sandbags:** Props must not leave holes in the Performance Field surface. Sand bags must be in a secondary container (bucket, double bagged, etc.) that will prevent sand from leaking.

Rule 8.07: Equipment Wheel and Turf Protection Standards

11. • **Pneumatic-Like Tires:** All instruments and equipment wheeled into the stadium from the entrance gate forward must have pneumatic-like tires. Wheels must be able to support the weight of the instrument/prop without creating damage to the field surface.

Rule 8.08: Prop Staging Height Restrictions

12. • **12-Foot Rigid Height Limit:** Staging (props, backdrops, screens, or similar objects) built and/or used by bands at CBA sanctioned events shall be limited to a maximum total height of twelve (12) feet, including wheels, platforms, and safety railings. Materials such as wood, metal, plastic, or aluminum are not permitted above the twelve-foot limit.

Rule 8.09: Prop Assembly, Timing, and USAFA Falcon Stadium Clearance

13. • **Staging Ingress:** A band may bring props into the stadium after the previous band has entered the field. Props must fit in the end zone curve without blocking entrance/exit gates.
14. • **Falcon Stadium 9'6" Tunnel Rule:** Any prop being used at Falcon Stadium (United States Air Force Academy) must not exceed 9 ft 6 in. to clear the tunnel entrance. Props up to the 12-ft limit may be erected in the tunnel after clearing the initial entrance.

15. • **Falcon Stadium Venue Update (2025/2026):** While the 9'6" tunnel restriction remains in the printed CBA rulebook, it has been rendered obsolete by stadium renovations completed prior to the 2025 season. The lower overhead crossbeam at the bottom of the Falcon Stadium tunnel was permanently removed, and these backdrop props cleared the tunnel entrance at their full upright height (~10.5 ft) without issue throughout the 2025 season.

Rule 9.07: Parent Field Access & Wristband Limits

16. • **25 Field Pass Wristbands:** Access to the field is restricted to Directors/Staff with CBA passes and parents assisting with props/front ensemble with designated Field Pass wristbands. Each band receives exactly 25 Field Pass wristbands upon check-in. Additional wristbands are not available.

1.6 Post-Use Teardown and Trailer Storage

- 1. Immediate De-Ballasting:** Remove all 15-lb sandbags from the 3/4" vertical wing posts and rear framing rail at the stadium exit chute immediately following performance exit. Never transport carts in trailers or across curbs with sandbags installed on the posts or frame.
- 2. Sandbag Trailer Staging:** Stack sandbags low over the trailer axles on rubber floor mats to maintain a low trailer center of gravity.
- 3. Cart Base Loading:** Roll empty cart bases into the front trailer nose wall standing upright (Figure 4). The 3D-printed Black ASA corner bumper braces prevent scuffing between adjacent carts.
- 4. Ratchet Strap Restraint:** Secure cart bases using heavy-duty E-track ratchet straps wrapped around the bottom 2x4 lumber rails (Figure 4). NEVER strap across the steel pipe frame or struts.
- 5. Frame & Banner Rack:** Store detached 10' x 8' steel frames with attached vinyl banners upright in protective padded storage racks (Figure 5).



Figure 4. Rolling cart bases strapped to front trailer nose wall for transit.



Figure 5. Stacked steel backdrop frames with installed vinyl banners in trailer storage rack.

PART OF THE ROLLING BACKDROP CONSTRUCTION AND FIELD OPERATIONS MANUAL

APPENDIX B

Construction Manual

Materials, cutting, cart assembly, ballast retention posts, instrument tote, steel frame, vinyl, and source lookup

This appendix is designed as a complete, independently printable work packet which can be distributed separately to fabrication volunteers and workshop build teams.

Audience: Fabrication volunteers and workshop build teams

Contents: Components; tools; 3D printed parts; 2x4 cart assembly; 3/4" ballast posts; HDX tote bag; steel frame; struts & pre-drilling; vinyl installation; Appendix B index

Print or carry: All pages labeled B-

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When separated from the master document, keep this cover as the first page of Appendix B.

Construction

Safety and work area

- 1. Personal Protective Equipment (PPE):** Always wear ANSI Z87.1 safety glasses and hearing protection when cutting lumber, operating metal saws, or drilling galvanized steel tubing.
- 2. Workspace Setup:** Provide a clean, flat 12 ft x 12 ft assembly floor for squaring the 96" x 44.5" cart base and 10' x 8' steel upright frame.

Materials and components

Component	Specification	Qty / Prop	Source
2x4 SPF / Doug Fir	2x4x8 ft framing lumber (96" rails, 44.5" joists)	8 pcs	Lumber yard
1/2" Plywood Decking	1/2 in. CDX / Sanded Plywood (Left & Right wings)	1 sheet	Lumber yard
3/4" Floor Flanges	3/4 in. Black Iron Threaded Floor Flanges	2 pcs	Index [16] / Home Depot
3/4" x 18" Iron Pipe	3/4 in. Dia. x 18 in. Threaded Black Iron Pipe (Posts)	2 pcs	Index [16] / Home Depot
Flange Carriage Bolts	1/4-20 x 1-1/2 in. Carriage Bolts, Washers, Lock Nuts	8 sets	Hardware store
HDX 14-Gal. Storage Tote	14-Gallon Tough Tote (Black/Yellow) - Instrument Carrier	1 pc	Index [9] / Home Depot
GRK #9 x 2-1/2" Screws	Star-drive R4 multi-purpose screws (Red markers)	~40 pcs	Index [5] / Home Depot
GRK #10 x 4" Screws	Self-countersinking flat head screws (Blue markers)	12 pcs	Index [6] / Home Depot
Swivel Casters	Heavy-duty turf-compatible non-locking swivel plate casters	4	Index [4] / Amazon
Fence Line Posts	1-5/8" Dia. x 10 ft 16-Ga. Galvanized Steel Posts (Black)	2	Index [1] / Home Depot
Fence Top Rails	1-5/8" Dia. x 8 ft Top Rails (Frame & Struts)	4	Home Depot
3-Way Corner Clamps	1-5/8" 3-Way Clamp Corner Elbow Brackets	4	Index [2] / Amazon
Fence Tension Bands	1-5/8" Galvanized Steel Tension Bands	4	Index [3] / Home Depot
Steel End Caps & U-Bolts	1-5/8" Steel Rail End Caps & Heavy-Duty U-Bolts	4 caps, 2 U-bolts	Index [7] / Amazon
15-lb Handle	Abccanopy 15-lb Heavy-Duty Sandbags w/	4–9 pcs (4–6 wing,	Index [13] /

Component	Specification	Qty / Prop	Source
Sandbags	Handle	3 rear rail)	Amazon
Sandbag Inner Liners	Heavy-Duty Plastic Sandbag Liner Bags (Double-Bag)	4–9 pcs (Double-Bag Compliance)	Index [14] / Amazon
70-lb Tube Sand (Reserve)	Sakrete 70-lb Traction Tube Sand + Contractor Bags (emergency reserve only)	1–2 pcs	Index [15] / Lowe's
Custom Vinyl Banner	Heavyweight outdoor scrim vinyl (dimensions TBD; flush straight cut, no hems/grommets)	1	Custom printed
Greenhouse Snap Clamps	1-1/4" Greenhouse Snap Clamps (sized for 1-5/8" fence pipe w/ vinyl)	14 pcs	Index [8] / Amazon
Double-Sided Tape	1-in. heavy-duty mounting tape (14x ~4-in. strips: 4 per long side, 3 per short side; TBD)	1 roll	Hardware supplier
Moving Blankets	Clean protective floor pads for scratch-free vinyl staging	3-4	Shop stock
Relief Hole Punch Tool	3/8-in. (10 mm) Rotary Leather Punch / Gasket Hole Punch	1 pc	Workshop stock

Tools and equipment

Required workshop tools and fixtures:

17. • Miter saw or circular saw with wood framing blade
18. • Metal chop saw or reciprocating saw with metal cutting blade
19. • Impact driver with Torx T-25 bit
20. • Cordless drill with 1/8-in. HSS / Cobalt drill bit rated for steel (pre-drilling 16-ga. tubing)
21. • 5/16-in. socket wrench, 7/16-in. wrench for 1/4-20 flange bolts, and open-end wrenches for U-bolt hardware
22. • Pipe wrench for threading 3/4-in. black iron pipe into floor flanges
23. • 25-ft tape measure, framing square, and speed square
24. • Isopropyl alcohol and microfiber rags for surface degreasing

3D-printed parts and fixtures

Print all fixtures in Black ASA or PETG for outdoor UV and heat stability. Do not use PLA.

Part / Tool	Qty	Purpose & Source File
Corner Bumper Braces	4	Protects 2x4 outer corners from scuffing; Index [10] (Hardware/Corner Brace.FCStd)
Drill Alignment Jig	1	Guides 1/8" pilot holes through 1-5/8" fence pipe; Index [11] (Hardware/Drill Alignment Jig.3mf)
Strut Bracket Marker	1	Locates base U-bolt & upper tension band positions; Index [12] (Hardware/Steel Support Strut Bracket Marker.FCStd)

Estimated fabrication cost breakdown

The following table itemizes estimated material costs per backdrop prop based on standard retail pricing (excluding custom printed vinyl):

Subsystem / Category	Items Included	Est. Cost
Cart Base Framing	8x 2x4 SPF lumber, 1/2 sheet 1/2" plywood, GRK screws, paint	\$98.00
Casters & Bumpers	4x Heavy-duty non-locking swivel casters, 4x ASA corner bumpers	\$51.00
Ballast Posts & Tote	2x 3/4" flanges & 18" pipes, carriage bolts, 1x HDX 14-gal tote	\$46.00
Upright Steel Frame	2x 10-ft 16-ga fence posts, 2x 8-ft top rails, 4x 3-way brackets, bands	\$121.00
Diagonal Struts & Mounts	1x 10-ft top rail (2 struts), 4x end caps, tension bands, 2x U-bolts	\$49.50
Vinyl Attachment Hardware	1-1/4" greenhouse snap clamps pack (14 pcs), 1" heavy-duty tape, pilot hardware	\$25.00
TOTAL PER PROP (EXCL. VINYL)	Hardware & lumber complete chassis	~\$390.00

Optional Rule 8.05 Ballast Pack: 4–9 Abccanopy 15-lb handle sandbags + inner double-bag liners + traction sand adds approximately +\$37.00 to \$55.00 per cart (scaled to the Tiered Ballasting Schedule in Section 1.4).

Frame assembly

Stage 1: Assemble the Rolling Cart Base

The rolling cart base is fabricated from 2x4 SPF/Doug Fir lumber according to the component layout and fastener map shown in Figure 1:

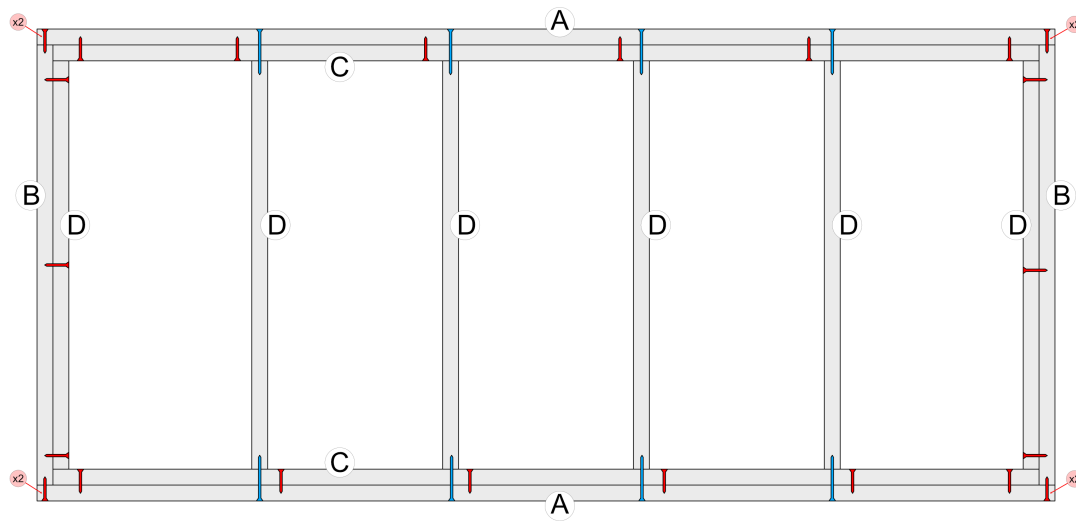


Figure 1. Top-down 2x4 rolling cart frame layout and screw fastening map.

Stage 1 Assembly Procedure:

- 1. Cut Lumber:** Cut outer longitudinal rails (A) to 96.0 inches and end perimeter rails (B) to 44.5 inches.
- 2. Square Perimeter:** Lay the four perimeter members flat on the shop floor. Measure opposite diagonals to confirm exact squareness.
- 3. Outer Corners:** Fasten each outer corner joint with two GRK #9 x 2-1/2 in. wood screws (red markers labeled 'x2').
- 4. Joist Layout:** Place six cross joists (D): two flush against the inner face of end rails (B), and four interior joists spaced to define the left decking wing, open tote bay, and right decking wing.
- 5. End Joists:** Secure each outer end joist (D) to end rail (B) with three GRK #9 x 2-1/2 in. screws.
- 6. Structural Screws:** Drive GRK #10 x 4 in. structural screws (blue markers) from outside rail (A) into the end grain of each cross joist (D) (two structural screws per joist).
- 7. Lamination:** Laminate inner reinforcement rails (C) to the inside faces of rails (A) using GRK #9 x 2-1/2 in. screws as indicated by the interior red markers.

Stage 2: Install Decking, Ballast Posts, Casters, and Instrument Tote

1. **Plywood Decking:** Fasten two 1/2-in. plywood panels across the left wing (outer rail B to 2nd joist D) and right wing (5th joist D to outer rail B) with 1-1/4 in. screws. Confirm the center bay between the 3rd and 4th joists remains completely clear to receive the HDX 14-gallon tough tote.
2. **Ballast Retention Post Flange Installation:** Locate the geometric center of each 1/2" plywood wing panel. Position a 3/4-in. black iron floor flange (Index [16]) and mark four bolt holes. Drill 1/4" clearance holes through the plywood. Secure each flange from the underside using 1/4-20 x 1-1/2 in. carriage bolts, flat washers, and nylon lock nuts. Thread a 3/4-in. x 18-in. black iron pipe into each flange and tighten firmly with a pipe wrench.
3. **Instrument Carrier Tote Installation:** Insert the central HDX 14-gallon tough storage tote (Index [9]) into the open center frame bay between the 3rd and 4th cross joists (D). The tote rim rests on the 2x4 framing to serve as the student instrument carrier during show operations. (Note: A future design revision is planned to accommodate larger instrument geometries).
4. **Painting:** Paint all cart bases and wood surfaces with exterior-grade semi-gloss black paint before final caster assembly (Figure 7).
5. **Swivel Casters:** Invert the cart base. Mount four heavy-duty swivel plate casters (Index [4]) directly beneath the corner lumber intersections using 5/16 in. x 1-1/2 in. structural timber screws with flat washers (Figure 9).
6. **Corner Bumpers:** Fasten four 3D-printed Corner Braces (Index [10]) over each outer corner with #8 pan-head screws to protect the wood cart from transport impacts.



Figure 7. Cart base frames painted black with plywood decking wings installed.



Figure 9. Swivel caster wheel assembly mounted beneath corner framing.

Operational Caster Fastening Standard: While early prototype builds utilized wing nuts as pictured in Figure 9, field operations standardized on standard hex nuts. Using standard hex nuts enables parent crew to rapidly remove and secure casters using an impact driver / cordless drill, significantly outperforming manual wing nut threading during teardowns.

Stage 3: Assemble the Upright Backdrop Frame

- 1. Upright Rectangle:** Join two 10-ft 16-gauge steel fence line posts and two 8-ft top rails using four 1-5/8 in. 3-way clamp corner elbow brackets (Index [2]). Bottom out all tube ends inside the clamp sockets and tighten clamp bolts evenly.
- 2. Base Attachment:** Stand the upright frame against the front rail (A) of the cart base. Anchor the bottom of each vertical post using 1-5/8 in. galvanized steel tension bands (Index [3]) through-bolted to the 2x4 lumber frame.

Stage 4: Fabricate and Install Diagonal Support Struts

Because 16-gauge galvanized steel is exceptionally hard, direct self-tapping screws will skate and dull. Follow this guided drilling protocol:

- 1. Strut Preparation:** Cut two diagonal strut lengths from 1-5/8 in. steel top rail and slide 1-5/8 in. steel end caps over each end.
- 2. Guided Pre-Drilling:** Clamp the 3D-printed Drill Alignment Jig (Index [11]) over the end cap pilot location. Using a 1/8-in. HSS / Cobalt drill bit rated for steel, drill clean pilot holes through the end cap and underlying pipe wall. Drive self-tapping screws into the pre-drilled holes to lock the end caps permanently.
- 3. Frame Mounting:** Mount upper tension bands on the vertical posts using the 3D-printed Strut Bracket Marker template (Index [12]). Bolt the upper strut end caps to the tension bands. Anchor the lower strut end caps to the rear 2x4 cart framing with heavy-duty steel U-bolts (Index [7]) as shown in Figure 8.



Figure 8. Completed rolling cart base and upright steel frame assembly.

Vinyl Installation

WORKING DRAFT — VINYL INSTALLATION PROCEDURE INCLUDED: Working draft procedure for vinyl installation using perimeter double-sided tape, hardware relief slits, and mechanical greenhouse snap clamps. Recommended vinyl dimensions (image size, bleed size, overall cut size) and tape strip quantities are provisional and subject to shop fit-up confirmation.

This section governs the preparation, alignment, dual-axis tensioning, relief slit cutting, and mechanical fastening of the custom printed vinyl coverings to completed backdrop upright frames. Installing vinyl on the 10 ft x 8 ft steel frame requires a clean staging area, careful bleed centering, razor relief cuts for protruding frame hardware, double-sided tape adhesion, and mechanical clamping with greenhouse snap clamps to prevent loosening under high stadium wind loads.

Recommended Vinyl Dimensions & Ordering Specifications

Custom printed backdrop banners must be ordered to exact dimensions to provide clean full-bleed framing around the outer perimeter of the steel pipe structure. The specifications below define the required production allowances:

Specification / Parameter	Engineering Requirement / Dimension
Nominal Upright Frame Size	8 ft Width × 10 ft Height (96.0 in. × 120.0 in. outer pipe perimeter)
Visible Graphic / Live Face	[TBD] in. Width × [TBD] in. Height (Front visible image area)
Perimeter Bleed Wrap Allowance	[TBD] in. per border (top, bottom, left, and right outer pipe wrap)
Overall Flat Cut Size	[TBD] in. Width × [TBD] in. Height (Including bleed allowance)
Substrate Material	13 oz or 18 oz heavy-duty outdoor matte scrim vinyl (UV and tear resistant)
Edge Perimeter Finish	Square raw-cut perimeter; DO NOT order hemmed borders or brass grommets, as raw edges wrap smoothly under snap clamps.

Required Materials & Fastening Hardware

- **Moving blankets:** Three to four clean, heavy-duty moving blankets or carpet runners spread across the shop floor to cushion the printed vinyl face against dust, grit, and concrete abrasion.
- **Custom vinyl banner:** One custom printed outdoor banner per backdrop (nominal 8 ft x 10 ft), rolled with print side facing inward. Unroll and inspect for print flaws and correct orientation prior to assembly.
- **Utility knife / razor:** Fresh, sharp razor knife for cutting precision relief slits around mounting brackets and tension bands.

- **Double-sided tape:** 1-in. wide heavy-duty outdoor mounting tape (Appendix B Table 1). Cut into fourteen (14) strips of approximately 4-in. length (~56 linear inches per frame). Application schedule: 4 strips on each vertical upright rail (8 total) and 3 strips on each cross rail (6 total). Tape quantities are provisional (TBD) and will be validated during build testing.
- **Greenhouse snap clamps:** Fourteen (14) 1-1/4 in. greenhouse clamp clips (Appendix B Index [8]). Snapped directly over the wrapped vinyl at every tape location (4 along each long vertical rail, 3 along each short cross rail). Note: While the frame uses 1-5/8 in. outer diameter fence post tubing, 1-1/4 in. greenhouse clamp clips provide the proper snug snap fit over the tubing, vinyl, and tape.
- **Surface preparation (Optional):** Isopropyl alcohol (IPA) and clean microfiber cloth. Degreasing the steel pipe is completely optional—the heavy-duty tape combined with mechanical greenhouse snap clamps provides more than enough holding power. A weaker initial adhesive bond to the bare steel pipe is actually preferable, as it significantly simplifies tape removal and cleanup later.
- **Personnel requirement:** Minimum two operators required; three to four recommended for synchronized dual-operator transverse tensioning and safe frame lifting.

Step-by-Step Vinyl Installation Procedure

Step 1: Staging and Surface Protection

Clear an unobstructed shop workspace measuring at least 12 ft x 14 ft. Thoroughly sweep and vacuum the area to remove metal chips, drill burrs, or grit that could puncture or scratch the printed vinyl face.

Lay clean moving blankets flat across the floor, overlapping edges to create a continuous, cushioned staging area larger than the overall banner flat cut size.

Carefully unroll the custom vinyl banner and lay it completely face down directly onto the moving blankets. Smooth the banner flat by hand and inspect the back side to confirm it is clean and free of trapped debris.

Step 2: Frame Placement and Bleed Alignment

With two operators lifting opposite ends, carry the assembled 10 ft x 8 ft steel upright frame onto the staging area and place it face down centered over the back side of the vinyl banner.

Measure the bleed wrap margin extending beyond the outer edge of the steel tubing along all four perimeter borders. Shift the frame incrementally until an equal reveal of bleed extends evenly on all four sides (top, bottom, left, and right outer rails).

Verify frame squareness by measuring crossed diagonals prior to applying tape or making relief cuts.

Step 3: Hardware Relief Slits

Before adhering tape or wrapping vinyl, use a fresh, sharp utility razor knife to make small, neat relief slits in the vinyl bleed wrap allowance to accommodate protruding frame hardware:

- **Long edges (Vertical Uprights):** Cut small slits directly aligned with the diagonal support strut attachment brackets (tension bands) on the left and right upright rails. These slits allow the strut brackets to protrude through the bleed wrap cleanly, preventing unsightly puckering, tenting, or bunching of the visible banner face.
- **Bottom edge (Cart-Mounting Interface):** Cut two small relief slits along the bottom rail bleed wrap corresponding to the base tension bands and mounting brackets that secure the upright frame to the wood cart base.

Step 3.1: Cutting Engineered Wind Relief Slits

To prevent violent wind flutter, reduce peak dynamic overturning impulses during stadium gusts, and protect greenhouse snap clamps from popping off under high wind loads, cut six (6) engineered inverted-U crescent wind relief flaps into the custom vinyl banner while it remains face-down on the cushioned moving blankets:

- **Mandatory Tear-Arrest Relief Holes:** CRITICAL FABRICATION RULE: NEVER cut wind relief slits purely with a utility razor knife leaving sharp 90° corners. Sharp razor corners create extreme stress concentration factors ($K_t \geq 3.0$) that will rapidly propagate into catastrophic tears across the entire banner during the first stadium wind gust. Before making any razor cuts, use a 3/8-in. (10 mm) rotary leather punch or clean gasket punch against a wooden backup block to punch two smooth, circular holes at the upper hinge corners of each planned flap.
- **Inverted "U" Crescent Flap Geometry:** Each flap measures 8.0 in. wide with a 5.0 to 6.0 in. downward curved drop. The top 8.0-in. span between the two punched relief holes remains uncut to serve as a flexible vinyl hinge. Using a fresh utility razor knife, cut a smooth downward curve starting from the bottom of one hole, curving down 5 to 6 inches, and terminating cleanly into the bottom of the second hole.
- **Aerodynamic Function & Audience Appearance:** Under calm conditions, gravity pulls the flap flush against the banner face. From audience stands 50 to 150 feet away, the narrow slit line is virtually invisible, preserving complete graphic visual continuity. Under forward or rear wind pressure, the flap swings open freely, venting dynamic pressure spikes and damping vortex-shedding flutter oscillations.
- **Upper-Venting Zone Layout (Row 1 & Row 2):** Because overturning moment increases directly with height above the turf ($M = F \cdot y$), venting pressure near the top of the prop is 2.5 times more effective at preventing tipping than venting near the bottom. Slits must be concentrated in the upper 40%–60% of the banner:

- **Row 1 (Upper Tier):** Located 8.0 ft (96 in.) above the bottom rail (2.0 ft below top rail). Center three flaps across the width at $X = 2.0$ ft (24 in.), $X = 4.0$ ft (48 in. centerline), and $X = 6.0$ ft (72 in.) from the left outer edge.

- **Row 2 (Mid-Upper Tier):** Located 5.5 ft (66 in.) above the bottom rail. Center three flaps across the width at $X = 2.0$ ft (24 in.), $X = 4.0$ ft (48 in. centerline), and $X = 6.0$ ft (72 in.) from the left outer edge.

- **Artwork & Graphic Clearance Rules:** Before punching or cutting, inspect the front face of the custom printed artwork. Adjust flap centerlines horizontally by ± 6 to 12 inches to place slits in solid background fields, skies, or low-contrast negative space. NEVER cut across student performer faces, school crests, band logos, or sharp typographic lettering.

Step 4: Perimeter Double-Sided Tape Application

Cut fourteen (14) strips of 1-in. heavy-duty double-sided tape, each approximately 4 inches in length.

Apply the tape strips to the rear-facing surface of the perimeter steel fence post tubing according to the following layout:

- **Long rails (Left and Right 10-ft Vertical Uprights):** Apply 4 strips along each upright (8 total). Crucially, position two of the strips immediately above and below the support strut bracket to anchor the vinyl around the bracket interface, and place two additional strips spaced along the remaining length of each upright.

- **Short end rails (Top and Bottom 8-ft Cross Rails):** Apply 3 strips evenly spaced along each rail (one centered and two approximately 6 inches inward from the corner brackets; 6 strips total).

Burnish each tape strip firmly onto the steel pipe surface, then peel and discard the protective release liners from all 14 strips.

Step 5: Long-Side Tensioning and Initial Wrap

Start along one designated long side (left vertical upright). Beginning at the support strut bracket, ease the bracket through its pre-cut relief slit, fold the vinyl bleed margin tightly around the steel pipe, and press it firmly into the 4 double-sided tape strips on the rear of the tube.

Move to the opposing long side (right vertical upright). Pull the vinyl firmly across the frame face to eliminate all slack and wrinkles. Ease the second support strut bracket through its relief slit, fold the vinyl edge tightly around the steel pipe, and press it securely into the 4 opposing tape strips.

Sight down the front face: confirm that longitudinal tension is smooth, uniform, and drum-tight, with no diagonal wrinkling or wave patterns.

Step 6: Short-Side Dual-Operator Tensioning and Wrap

Station operators at opposite short ends (top cross rail and bottom cross rail).

Working simultaneously, both operators firmly grasp the vinyl bleed margin on opposite ends and pull outward in opposing directions away from the center. This synchronized outward pull establishes balanced transverse tension across the entire frame without racking the structure out of square.

While maintaining continuous outward tension, wrap the top vinyl edge over the top cross rail and press into the 3 tape strips. At the bottom rail, guide the cart-mounting base brackets through their two pre-cut relief slits, wrap the vinyl edge tightly around the bottom pipe, and press into the 3 bottom tape strips.

Neatly fold excess vinyl corner tabs around the 3-way corner elbow brackets, flattening the material to maintain clean, square corners without bulky bunches.

Step 7: Greenhouse Snap Clamp Installation

Install the 1-1/4 in. greenhouse snap clamps directly over the wrapped vinyl at every tape location, pressing each clamp firmly over the pipe until it snaps into full mechanical lock:

- **Long rails (Vertical Uprights):** Install 4 snap clamps on each upright (8 clamps total). Crucially, position two of the snap clamps immediately above and below each support strut bracket to firmly sandwich and hold the bracket in place, preventing bracket slip, flutter, or vinyl detachment under wind loads.
- **Short end rails (Top and Bottom Cross Rails):** Install 3 snap clamps on the top rail and 3 snap clamps on the bottom rail (6 clamps total).

The 1-1/4 in. greenhouse snap clamps compress the vinyl tightly against the tape and 1-5/8 in. steel fence post tubing, providing heavy-duty mechanical clamping that prevents edge detachment under stadium wind gusts or rough transport handling.

Confirm that no snap clamp interferes with the cart-mounting base brackets or corner elbow clamp bolts.

Step 8: Final Tension Inspection and Frame Re-Mounting

With three to four operators, carefully lift the completed backdrop frame upright off the moving blankets.

Inspect the front vinyl face in bright lighting: verify that the graphic is drum-tight, square, free of wrinkles or sagging, and cleanly fitted around all protruding bracket slits.

Verify that all 14 greenhouse snap clamps are fully seated and firmly gripping the tubing over each tape strip.

Re-secure the vertical upright posts into the cart base tension bands and re-connect the diagonal support struts. Confirm that the wrapped vinyl and snap clamps do not bind or impede strut attachment.



Figure 6. Installation team applying carpet tape and securing vinyl banner with pipe snap clamps.

Vinyl Removal & End-of-Season Teardown Protocol

CAUTION — Temperature Requirement for Vinyl Removal: Do not attempt removal of the vinyl or adhesive tape unless the ambient temperature is above 80°F (27°C). Attempting to peel the vinyl at lower temperatures will cause the material to tear and permanently damage the custom banner face. In cooler weather, warm the shop or gently heat the taped perimeter with a heat gun on low before peeling.

When taking down backdrops or preparing frames for graphic replacement:

- **Recommended timing (Defer to Band Camp):** It is usually much easier to leave the vinyl on the frames over the winter and wait until summer band camp of the following season before attempting removal. Cold late-fall temperatures following competition season make adhesive stiff and brittle, drastically increasing tear risk.
- **Solar heating advantage:** The warmer it is outside, the better. Staging the backdrops outdoors in direct sunlight for just a few minutes prior to peeling warms the steel tubing and softens the tape adhesive, allowing the vinyl and tape to release cleanly and effortlessly with minimal pull resistance.
- **Remove greenhouse snap clamps:** Carefully pry off all 14 greenhouse snap clamps from the perimeter tubing.

- **Peel slowly at shallow angle:** Starting at one corner, slowly peel the vinyl wrap back at a shallow angle. Warm adhesive releases cleanly from the bare steel pipe without pulling the scrim or ink layer.
- **Tape stripping & cleanup:** Peel remaining tape strips from the steel pipe. Because the pipe was not aggressively degreased during initial assembly, tape strips will pull off cleanly with minimal adhesive residue.

Final construction inspection

Verify before releasing prop for field use:

- ❑ **Cart Base Squareness:** Opposite diagonal measurements within 1/8 in.
- ❑ **Ballast Posts:** 3/4" flanges through-bolted securely to plywood; 18" pipes tight without wobble.
- ❑ **Instrument Tote Bay:** HDX 14-gallon tote drops smoothly into center frame bay.
- ❑ **Fasteners:** All GRK red/blue screws fully countersunk without wood splitting.
- ❑ **Casters:** Smooth 360° swivel action; hex nuts tightened securely; wheels roll freely without wobble (non-locking agility design).
- ❑ **Upright Frame:** All 3-way corner clamp bolts torqued securely.
- ❑ **Support Struts:** End cap screws and base U-bolts tight with zero play.
- ❑ **Vinyl:** Banner taut, wrinkle-free, and locked with snap clamps along all four sides.

Appendix B Index

Purchase sources

- [1] 1-5/8 in. Dia. x 10 ft Steel Fence Line Posts - Home Depot
<https://www.homedepot.com/p/Everbilt-1-5-8-in-Dia-x-8-ft-16-Gauge-Galvanized-Steel-Chain-Link-Fence-Line-Post-328923DPTSEB/312373067>
- [2] 1-5/8 in. 3-Way Clamp Corner Elbow Brackets - Amazon
<https://www.amazon.com/dp/B0CD7QT6L6>
- [3] 1-5/8 in. Galvanized Steel Tension Bands - Home Depot
<https://www.homedepot.com/p/Everbilt-1-5-8-in-Galvanized-Steel-Chain-Link-Fence-Tension-Band-328521EB/312373099>
- [4] Heavy-Duty Swivel Plate Casters - Amazon
<https://www.amazon.com/dp/B0CPXFJCL>
- [5] GRK #9 x 2-1/2 in. Star Drive R4 Multi-Purpose Screws - Home Depot
<https://www.homedepot.com/p/GRK-Fasteners-9-x-2-1-2-in-Star-Drive-Torx-Bugle-Head-R4-Multi-Purpose-Wood-Screw-300-Pack-100101/203533402>
- [6] GRK #10 x 4 in. R4 Self-Countersinking Flat-Head Screws - Home Depot
<https://www.homedepot.com/p/GRK-Fasteners-10-x-4-in-R4-Self-Countersinking-Flat-Head-Multi-Purpose-Screw-50-per-Pack-103141/203525231>
- [7] Heavy-Duty Steel U-Bolts - Amazon
<https://www.amazon.com/dp/B09L41JFMS>
- [8] Greenhouse snap clamps (1-1/4 in.) - Amazon
<https://www.amazon.com/dp/B0CDZP8YVF>
- [9] HDX 14-Gallon Tough Storage Tote (Black/Yellow) - Home Depot
<https://www.homedepot.com/p/HDX-14-Gal-Tough-Storage-Tote-in-Black-with-Yellow-Lid-999-14G-HDX/328027053>
- [13] Abccanopy 15-lb Heavy-Duty Sandbags with Handle (4-Pack) - Amazon
<https://www.amazon.com/dp/B0DFVZDVK>
- [14] Heavy-Duty Plastic Sandbag Liner Bags (Double-Bag Compliance) - Amazon
<https://www.amazon.com/dp/B0BG3F5XVS>
- [15] Sakrete 70-lb Traction Tube Sand - Lowe's
<https://www.lowes.com/pd/Sakrete-0-07-cu-ft-70-lb-Traction-Sand/5015728687>
- [16] 3/4 in. Black Iron Floor Flange and 18 in. Threaded Pipe - Home Depot
<https://www.homedepot.com/p/The-Plumber-s-Choice-3-4-in-Black-Malleable-Iron-Floor-Flange-5-Pack-FBNF034-5/308967916>

Digital part files & governing rules

[10] 3D Printed Corner Bumper Brace - FreeCAD Model

<https://github.com/ericrowe/music/blob/main/PCHSMB/Backdrop/Hardware/Corner%20Brace.FCStd>

[11] 3D Printed 1-5/8" Pipe Drill Alignment Jig - Slicer 3MF

<https://github.com/ericrowe/music/blob/main/PCHSMB/Backdrop/Hardware/Drill%20Alignment%20Jig.3mf>

[12] 3D Printed Support Strut Bracket Marker - FreeCAD Model

<https://github.com/ericrowe/music/blob/main/PCHSMB/Backdrop/Hardware/Steel%20Support%20Strut%20Bracket%20Marker.FCStd>

[17] 2026 Colorado Bandmasters Association (CBA) Marching Band Rulebook (PDF)

https://ebf5c7e8-3869-4325-bd98-ce10f57d7545.filesusr.com/ugd/83f67e_cb5fe6efdbdb4ce98f619580f694bc94.pdf